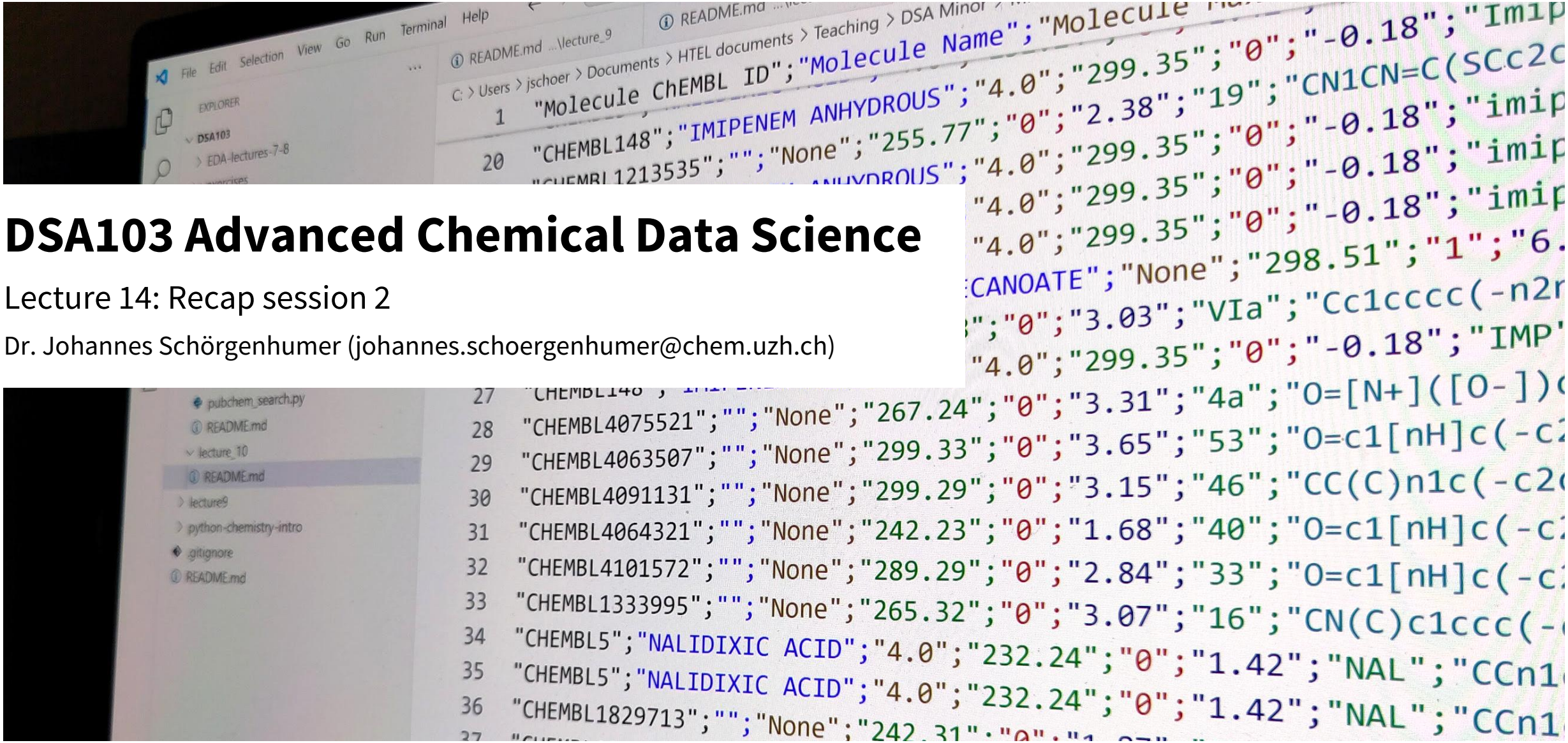
Universität Zürich[UZH]

Department of Chemistry



DSA103 Advanced Chemical Data Science

Lecture 14: Recap session 2

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Content overview

— Wrap up of RDKit workshop

— Exercises and quiz time

— Q&A

Activity: RDKit Workshop

1) Load and explore chemical datasets (Delaney solubility dataset)
2) Compute RDKit descriptors
3) Build regression model for molecular property prediction (data quality assessment)
4) Calculate fingerprints
5) Measure structural similarity (Tanimoto)
6) Cluster molecules by structure
7) Identify and query substructures using SMARTS patterns

Exercise: Distance calculation

Calculate the C-C bond length of the alkyne in yne.xyz

Quiz time!

What does **‘type(3/2)’** return?

Quiz time!

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Quiz time!

What does **immutable** mean, and which data type comes to mind there?

Quiz time!

Consider a list '**values=[10, 20, 30, 40, 50, 60, 70]**'. What is the output of '**print(values[1:4])**'?

Quiz time!

Consider '**a = np.array([[1, 2, 3], [4, 5, 6]])**'. What does '**print(a.shape)**' return?

Quiz time!

What are **aggregates** in numpy/pandas?

Quiz time!

Which pandas call gives you **column names, dtypes and non-null counts** of a DataFrame ‘df’ in one go?

Quiz time!

What are differences between ‘**df.loc**’ and ‘**df.iloc**’?

Quiz time!

In a plate-reader experiment, you record the treatment group (control, low dose, high dose) for each well. Which data type is this?

a) Continuous
b) Ordinal categorical
c) Nominal categorical
d) Binary

Quiz time!

A column of qPCR Ct values contains the entry “Undetermined” for several wells. What is the best first step?

a) Replace it with 0
b) Delete all rows containing it silently
c) Investigate why (no amplification vs. failed reaction), then encode as missing (NA) and document the decision
d) Replace it with the column mean

Quiz time!

You open a CSV in Excel and the gene name “SEPT2” turns into a date. What is the most robust way to avoid this?

a) Manually fix the dates afterward
b) Import the column explicitly as text, or work in code (pandas/R) with defined column types
c) Rename all genes to lowercase
d) Save the file as .xlsx instead of .csv

Quiz time!

What is the most appropriate way to report results from an experiment with n = 3 biological replicates per group?

a) Bar chart of means with SEM only
b) Individual data points plus summary statistics, stating n and what a replicate is
c) Only the p-value
d) Means with no measure of variability

Quiz time!

Which practice best supports reproducible data reporting in a lab?

a) Overwriting the raw data file with the cleaned version to save space
b) Keeping raw data read-only and doing all cleaning in a documented, scripted pipeline
c) Cleaning by hand in a spreadsheet and noting changes from memory
d) Sharing only the final figures

Q&A

Content?

Organisation?

Feedback?